

SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105**

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS

Course Code:	DSA0501
Course Title:	Query Processing for Data Science
CO Assessed:	CO1
Assessment:	Tool 2 – Application Based Questions
Weightage:	_____
Total Marks:	20 Marks

CO1 Statement: _____

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Section/Batch: A/2023_____ Date of Submission: 27/7/2026_____

Faculty In-charge: Dr.Veena Devi Submission Mode: Individual

Code of Conduct

I Surya JK certify that this submission is my original work and have adhered to all academic integrity guidelines.

Surya JK

Signature:

Criteria	Marks
Understanding of the Problem	4
Application of Concepts	4
Program Design / Algorithm	4
Correctness of Solution & Output	4
Explanation / Interpretation of Results	4
Total	20

CO1 – AT2 – Application Based Questions

Sample Questions

Application Based Questions

Electricity Consumption Analysis (10 Marks)

An electricity board receives monthly power consumption data from different cities in CSV format. Some records contain invalid meter readings and missing consumer IDs. Write a Python program to identify incorrect records, clean the dataset, and generate a summary showing the total electricity consumption for each city.

Solution:

```
import pandas as pd

def analyze_electricity_consumption(input_file: str,
    summary_output_file: str): # 1. Read CSV file
    df = pd.read_csv(input_file)

    # 2. Remove records with missing
    Consumer IDs df =
    df.dropna(subset=['consumer_id'])

    # 3. Clean invalid meter readings (ensure consumption is
    numeric and >= 0) df['units_consumed'] =
    pd.to_numeric(df['units_consumed'], errors='coerce') valid_df
    = df[df['units_consumed'] >= 0].copy()
```

Employee Data Integration (10 Marks)

A company maintains employee details in a CSV file and department information in a JSON file. Both files contain a common Department ID field. Write a Python program to read the two files, combine the employee and department information, identify records with missing department details, and generate a consolidated dataset that can be stored in a relational database.

Solution:

```
import
pandas as
pd import
json

def merge_employee_department(csv_file: str, json_file: str,
    output_file: str): # 1. Read CSV employee data
    emp_df = pd.read_csv(csv_file)

    # 2. Read JSON
    department data with
```

```

open(json_file, 'r') as
f:
    dept_data = json.load(f)
dept_df = pd.DataFrame(dept_data)

# 3. Merge datasets on Department ID (Left Join to keep all
employees) merged_df = pd.merge(emp_df, dept_df,
on='department_id', how='left')

# 4. Identify records with missing department
details missing_dept =
merged_df[merged_df['department_name'].isna()]
print(f"Found {len(missing_dept)} employees with
unassigned/missing departments.")

# 5. Fill NaN values in department fields before database export
merged_df['department_name'] =
merged_df['department_name'].fillna('Unassigned')

# Save to CSV (Ready for relational DB import)
merged_df.to_csv(output_file, index=False)
print(f"Consolidated dataset saved to
{output_file}")

# Example Usage:
# merge_employee_department('employees.csv', 'departments.json',
'consolidated_employees.csv')

```

Library Database Preparation (10 Marks)

A library receives book details in JSON format from different publishers. The records include book ID, title, author, publisher, and year of publication. Before storing the information in the library database, the data must conform to the database schema. Write a Python program to read the JSON file, validate the required fields, detect missing or inconsistent information, and produce a cleaned dataset ready for database insertion.

Solution:

```
import json

required_fields = ["book_id", "title", "author", "publisher", "year"]

with open("books.json", "r") as file:

    books = json.load(file)

    cleaned_data = []

    for book in books:

        valid = True

        for field in required_fields:

            if field not in book or str(book[field]).strip() == "":

                print(f'Missing {field} in record: {book}')

                valid = False

        if valid:

            if not isinstance(book["year"], int):

                print(f'Invalid year in record: {book}')

                valid = False

        if valid:

            cleaned_data.append(book)

    with open("cleaned_books.json", "w") as file:

        json.dump(cleaned_data, file, indent=4)

    print("Cleaning completed.")

    print("Valid records:", len(cleaned_data))
```

